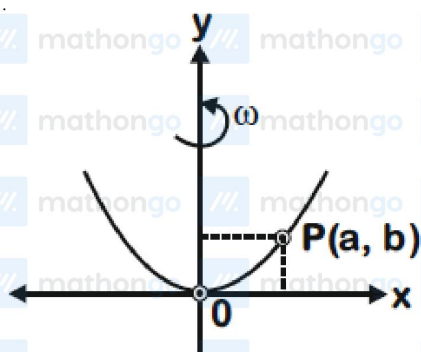


Q1 JEE Main 2020 - 2 September (Morning)

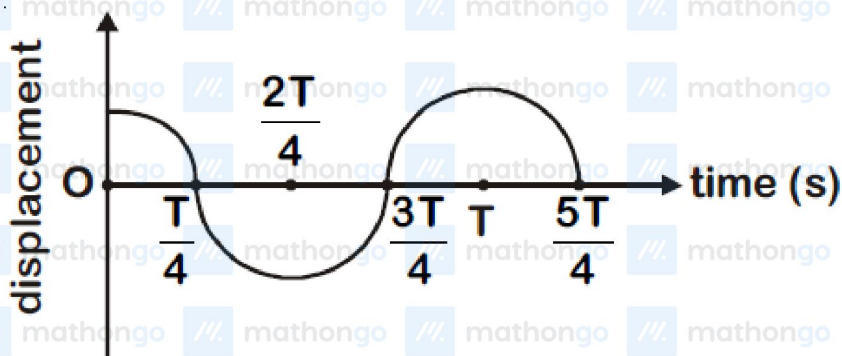
A bead of mass m stays at point $P(a, b)$ on a wire bent in the shape of a parabola $y = 4Cx^2$ and rotating with angular speed ω (see figure). The value of ω is (neglect friction)



- (A) $2\sqrt{2gC}$
 (B) $2\sqrt{gC}$
 (C) $\sqrt{\frac{2g}{C}}$
 (D) $\sqrt{\frac{2gC}{ab}}$

Q2 JEE Main 2020 - 2 September (Evening)

The displacement time graph of a particle executing S.H.M is given in figure (sketch is schematic and not to scale)



Which of the following statements is/are true for this motion?

- (A) The force is zero at $t = \frac{3T}{4}$

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(B) The acceleration is maximum at $t = T$

(C) The speed is maximum at $t = \frac{T}{4}$

(D) The P.E. is equal to K.E. of the oscillation at

$$t = \frac{T}{2}$$

(A) (B), (C) and (D)

(B) (A) and (D)

(C) (A), (B) and (C)

(D) (A), (B) and (D)

Q3 JEE Main 2020 - 3 September (Evening)

A block of mass m attached to a massless spring is performing oscillatory motion of amplitude 'A' on a frictionless horizontal plane. If half of the mass of the block breaks off when it is passing through its equilibrium point, the amplitude of oscillation for the remaining system becomes fA . The value of f is

(A) $\frac{1}{\sqrt{2}}$

(B) $\sqrt{2}$

(C) 1

(D) $\frac{1}{2}$

Q4 JEE Main 2020 - 5 September (Evening)

A ring is hung on a nail. It can oscillate, without slipping or sliding (i) in its plane with a time period T_1 and, (ii) back and forth in a direction perpendicular to its plane, with a period T_2 . The ratio $\frac{T_1}{T_2}$ will be

(A) $\frac{\sqrt{2}}{3}$

(B) $\frac{2}{\sqrt{3}}$

(C) $\frac{2}{3}$

(D) $\frac{3}{\sqrt{2}}$

Q5 JEE Main 2020 - 6 September (Evening)

When a particle of mass m is attached to a vertical spring of spring constant k and released, its motion is described by $y(t) = y_0 \sin^2 \omega t$, where y is measured from the lower end of unstretched spring.

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Then ω is

(A) $\sqrt{\frac{2g}{y_0}}$

(B) $\frac{1}{2} \sqrt{\frac{g}{y_0}}$

(C) $\sqrt{\frac{g}{2y_0}}$

(D) $\sqrt{\frac{g}{y_0}}$

SHM

Questions with Answer Keys

JEE Main 2020 Chapterwise

Physics

Answer Key

Q1 (A)

Q2 (C)

Q3 (A)

Q4 (B)

Q5 (C)

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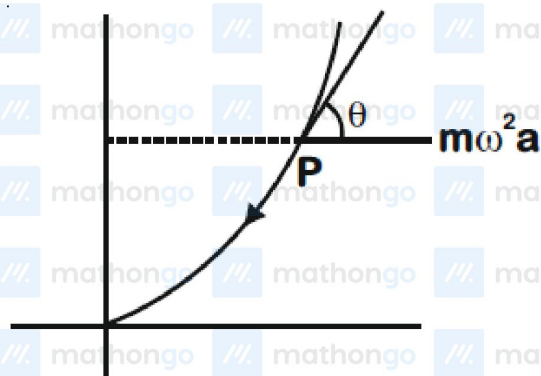
SHM

Hints & Solutions

JEE Main 2020 Chapterwise

Physics

Q1



$$y = 4Cx^2$$

$$\Rightarrow \frac{dy}{dx} = \tan \theta = 8Cx$$

$$\text{At } P, \tan \theta = 8Ca$$

For steady circular motion

$$mg \sin \theta = m\omega^2 a \cos \theta$$

$$\Rightarrow \tan \theta = \frac{\omega^2 a}{g}$$

$$\Rightarrow 8Ca \times g = \omega^2 \times a$$

$$\Rightarrow \omega = \sqrt{8gC}$$

$$\Rightarrow \omega = 2\sqrt{2gC}$$

Q2

At $t = \frac{3T}{4}$, particle is at mean position

$\Rightarrow$ Force = 0

At $t = T$, particle is at extreme position

$\Rightarrow$ Acceleration is maximum

At $t = \frac{T}{4}$, particle is at mean position

$\Rightarrow$ Speed is maximum

Q3

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$$A'\omega = A_0\omega_0$$

$$\Rightarrow A' = A_0 \frac{\omega_0}{\omega}$$

$$A' = \frac{A_0}{\sqrt{2}}$$

Q4

$$T = 2\pi \sqrt{\frac{1}{Mgd}}$$

$$\frac{T_1}{T_2} = \sqrt{\frac{l_1}{l_2}}$$

$$I_1 = 2MR^2, I_2 = \frac{MR^2}{2} + MR^2 = \frac{3}{2}MR^2$$

$$\Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{4}{3}} = \frac{2}{\sqrt{3}}$$

Q5

$$y(t) = \frac{y_0}{2} (1 - \cos 2\omega t)$$

$$\frac{Mg}{K} = \frac{y_0}{2} [\text{Amplitude}]$$

$$2\omega = \sqrt{\frac{K}{m}}$$

$$\omega = \frac{1}{2} \sqrt{\frac{K}{m}} = \frac{1}{2} \sqrt{\frac{2g}{y_0}} = \sqrt{\frac{g}{2y_0}}$$